STA2005S - Linear Models

Course Outline

2026

STA2005S is an introduction to statistical modelling, and to the design and analysis of experiments. The course is both theoretical and applied: we cover the theory and mathematics underlying the estimation and inference, and also the application of the theory to real data sets.

There are two sections:

- Applied Regression Analysis (30 lectures)
- Design and Analysis of Experiments (30 lectures)

The course counts 24 NQF credits, which amounts to a total workload of roughly 240 hours! This includes lectures, tutorials, exam preparation, everything. But it is still a lot of time you will need to invest.

Instructors

Mr Miguel Rodo	miguel.rodo@uct.ac.za	PD Hahn 5.52	
Dr Birgit Erni	birgit.erni@uct.ac.za	PD Hahn 6.64	Course Convenor

Course Entry Requirements

MAM1032S, STA2004F

MAM2011F (Linear Algebra) highly recommended.

Lecture Period and Venue

8am Monday to Friday in PD Hahn 2

Consultation Times

to be confirmed

Course Administrator

Ms Nodumo Maqubela, nodumo.maqubela@uct.ac.za

Please contact Nodumo for any administrative issues. Nodumo deals with several courses, so please mention that you are a STA2005S student in any email communication.

Course Content

STA2005S is an introduction to statistical modelling, the theory of linear statistical models and the principles of experimental design. Students are introduced to statistical software and practical data analysis through weekly computer practicals and the exposure to many data sets.

Regression

The multivariate normal distribution; quadratic forms; the linear model; maximum likelihood; estimates of parameters in the linear model; the Gauss-Markov theorem; variable selection procedures; residual analysis and model diagnostics; bootstrapping; principal components.

Design and Analysis of Experiments

Introduction to causal inference as a bridge between regression / observational studies and experiments, basic design principles, basic experimental designs (completely randomised designs, randomised block designs, Latin square designs), factorial experiments, analysis of variance, the problem of multiple comparisons, power and sample size calculations, introduction to random effects and repeated measures, non-parametric tests.

Introduction to R

In this course you will be introduced to a statistical software package called R. R is open-source; it is widely used by researchers, postgraduate students, and professionals. You can download R from http://cran.r-project.org/. See Amathuba (R Resources) for R resources, books, online tutorials and help.

The computing part of statistics is becoming increasingly important. Even if AI can do a lot of this now, it is still important that you understand the what it is doing, what it is doing wrong, and can guide and correct it. During your Honours year (and in the workplace) most of your time will be spent on instructing computers to do statistics for you. It is important to develop these skills. Programming and computer simulations are also an excellent way to develop a deeper understanding of the statistical methods.

For the two assignments in this course you will be required to use R for data analysis and data exploration and to write your reports.

The lab test later during the course will require that you are able to work with R.

Tutorials & Computer Practicals

Each student should attend one one-hour tutorial per week. During these sessions you have the opportunity to work on a set of problems together with other students, and to ask tutors for help.

In addition to the weekly tutorials, each student should attend one one-hour computer practical per week. During these prac sessions you will use R (Computer Labs). In addition to learning to use R for doing statistics, and learning to interpret the output, the practicals will involve computer simulations and explorations. These are useful for understanding concepts in a different way and more deeply, to explore concepts visually, and to develop your statistical computing skills.

- Tutorials and computer practicals are meant to help you learn and obtain a deeper understanding of the concepts. Tutorial and prac questions will not necessarily resemble test and exam questions but rather are constructed to help you work through concepts and build on what you already know.

Sign up for tutorials and computer pracs during week 1 on Amathuba.

Course Notes and Material

Course notes are made available electronically, and hard copies are available from reception in the Department of Statistical Sciences.

Assessments

1. There will be two **class tests**, one for each section.
    - **Test 1:** Tuesday 2 September 6-8pm, JD LT2 & James LT 4A
    - **Test 2:** Monday 13 October 6-8pm, JD LT2 & James LT 4A
2. **Practical Test**. This will run in the computer lab: you will need to use R for specified tasks. Internet access will be blocked. Date and venue to be confirmed.
3. There are two **assignments**, one for each module (Regression and Experimental Design). Details to be confirmed. The assignments will require proficiency in R. It is therefore important that you start to learn and use R from week one.
4. One final **exam** in November counts 70% of the final mark.

Missed Class Tests

Should you miss a test due to medical reasons, a valid medical certificate should be handed in to the course administrator within four working days of the missed class test. In the case of missed tests due to medical reasons, your class record will be calculated as the average of the other tests and assignments, and for your final mark your exam mark will make up for the proportion of the missed class test/s.

Course Marks

Your **final mark** will be calculated as: 0.3 class mark + 0.7 exam mark

Your **class mark** will be calculated with as follow with the following weights:

25% Test 1, 25% Test 2, 20% Assignment 1, 10% Practical Test, 20% Assignment 2.

DP Requirements

Satisfactory completion of **both** assignments, subminimum of 40% for average assignment mark.

Class record of at least 35%.

Course Schedule

STA2005S Course Schedule 2026

WEEK	START DATE	SECTION	Assessments	TUTORIAL
Week 1	27 July	R1		R Intro + R1
Week 2	3 Aug	R2		R2
Week 3 10th: Women's Day	10 Aug	R3		R3
Week 4	17 Aug	R4		R4
Week 5	24 Aug	R5		R5
Week 6	31 Aug	R6	Test 1: 1 Sep, Assign 1: TBC	R6
Vac	7 Sept			
Week 7	14 Sept	ED1: Design		ED1
Week 8 24th: Heritage	21 Sept	ED2: Linear Model		ED2
Week 9	28 Sept	ED3: ANOVA		ED3
Week 10	5 Oct	ED4: Contrasts		ED4
Week 11	12 Oct	ED5: RBD & LSD	Test 2: 12 Oct	ED5
Week 12	19 Oct	ED6: Factorial Experiments	Assignment 2: 19 Oct	ED6
Week 13	26 – 27 Oct			
Consolidation	28 Oct – 3 Nov			
Exams	4 Nov – 21 Nov			

Academic Honesty and Integrity

Please familiarise yourselves with UCT's plagiarism rules: https://uct.ac.za/administration/policies. Under no circumstances will plagiarism be tolerated.

Statement on the use of AI

While generative AI can provide useful tools to enhance some aspects of academic work, the University is deeply concerned that inappropriate AI use can undermine student learning and lead to academic dishonesty.

There are additional concerns that AI-generated work may be inaccurate, incomplete and biased. We therefore take the position that submitted work should not be produced using any generative AI tools – including use of AI to identify readings, summarise readings, generate code structure, and to write or edit text – except with the express written permission of the module lecturer that clearly delineates acceptable and appropriate AI use as per their specific course and assignments. Ideally, your task as a student is to learn and develop core academic skills independent of AI.

Your assignments will make clear what you are allowed to use, and how you should declare your use of AI.

In the event of interruption to face-to-face teaching

If face-to-face lectures or practicals are disrupted and/or access to campus is restricted, teaching and learning will continue online, with live online lectures delivered via Microsoft Teams wherever possible. All live lectures will be recorded as well.

The course convenor will make announcements on Amathuba about changes to the mode or timing of any activities associated with the course (lectures, pracs, tuts, tests, etc.). Students without internet access at home should take note of Eduroam locations (https://eduroam.ac.za/) and other places where free Wi-Fi is available.

Course Evaluations

At the end of the semester you have the opportunity to evaluate the course. Please use this opportunity to give constructive feedback so that we can improve the course.

Supplementary Exams

The deferred and supplementary exam will be in January 2027 (early January).

A supplementary exam is given to students who have obtained a final mark between 45 and 49 for STA2005S. In the case of a supplementary exam, the final mark (supp mark) equals the mark obtained for the supplementary exam (no class record).